Lecture 21

The abelian Higgs mechanism

A Maxwell field has two propagating polarisations. A massive vector has three. Coupling the two real components of a complex scalar to the gauge potential supplies the additional mode without changing the total count. We derive this statement near a nonzero scalar vacuum.

Objectives. *We use local amplitude and phase coordinates to split the quadratic action into a radial scalar and a massive vector. We count the vector's independent components and examine the zero-coupling limit and the obstruction at a vortex.*

By the end of this lecture you will be able to:

- Derive both masses with the course's scalar normalisation.
- Prove the Proca constraint and identify its three polarisations.
- Distinguish a local unitary gauge from a gauge smooth through a vortex zero.

21.1 The charged scalar model

Work on Minkowski spacetime $\mathbb{R}^{1+3}$, with smooth classical fields. Let $q > 0$, $v > 0$ and $\lambda > 0$ be constants. The complex field is $\Phi = \phi_1 + \mathrm{i}\phi_2$, as in Lecture 17, without a factor $1/\sqrt{2}$.

Definition 21.1 (Abelian Higgs model)**.** The *abelian Higgs model* has density

$$L = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\overline{D_\mu\Phi}D^\mu\Phi - \frac{\lambda}{8}(|\Phi|^2 - v^2)^2, \qquad D_\mu = \partial_\mu - \mathrm{i}q(A_{\mathrm{em}})_\mu, \tag{21.1}$$

where $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$. Its gauge transformations are

$$\Phi \longmapsto e^{\mathrm{i}q\chi}\Phi, \qquad A_{\mathrm{em}} \longmapsto A_{\mathrm{em}} + \mathrm{d}\chi. \tag{21.2}$$

The product rule gives $D'_\mu(e^{\mathrm{i}q\chi}\Phi) = e^{\mathrm{i}q\chi}D_\mu\Phi$. Together with $\mathrm{d}^2\chi = 0$, this proves gauge invariance of the density. The scalar potential has the vacuum circle $|\Phi| = v$. We expand near the constant vacuum $\Phi = v$, $A_{\mathrm{em}} = 0$.

21.2 Local amplitude and phase

On a sufficiently small patch where $\Phi \neq 0$, choose a smooth real phase and write

$$\Phi = (v + h)e^{\mathrm{i}\theta}, \qquad |h| < v. \tag{21.3}$$

Here $v + h$ is positive. The derivative is

$$D_\mu\Phi = e^{\mathrm{i}\theta}\left[\partial_\mu h + \mathrm{i}(v + h)(\partial_\mu\theta - q(A_{\mathrm{em}})_\mu)\right].$$

The mixed terms cancel on multiplication by the complex conjugate. The density is therefore exactly

$$\begin{aligned} L = &-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\partial_\mu h \partial^\mu h \\ &+ \frac{1}{2}(v+h)^2(\partial_\mu\theta - q(A_{\mathrm{em}})_\mu)(\partial^\mu\theta - q(A_{\mathrm{em}})^\mu) - \frac{\lambda}{8}(2vh+h^2)^2. \end{aligned} \tag{21.4}$$

The phase transforms as $\theta \mapsto \theta + q\chi$. Thus $\mathrm{d}\theta - qA_{\mathrm{em}}$ is gauge invariant.

Definition 21.2 (Gauge-invariant vector and unitary gauge)**.** On this phase patch, the *Stueckelberg vector* is the one-form

$$B = A_{\mathrm{em}} - q^{-1}\mathrm{d}\theta. \tag{21.5}$$

The *unitary gauge* is a choice with $\theta = 0$ and Φ positive real.

The function $\chi = -\theta/q$ reaches unitary gauge on the patch. The vector B is unchanged by gauge transformations, and $\mathrm{d}B = \mathrm{d}A_{\mathrm{em}}$ there. With $F_B = \mathrm{d}B$, the exact local density becomes

$$L = -\frac{1}{4}(F_B)_{\mu\nu}(F_B)^{\mu\nu} + \frac{1}{2}\partial_\mu h\partial^\mu h + \frac{1}{2}q^2(v+h)^2 B_\mu B^\mu - \frac{\lambda}{8}(2vh+h^2)^2. \tag{21.6}$$

The phase has disappeared from this gauge-invariant expression.

Proposition 21.3 (Quadratic Higgs split)**.** *Near the chosen vacuum, the terms of degree two in h, B and their derivatives are*

$$\begin{gathered} L^{(2)} = \frac{1}{2}\partial_\mu h\partial^\mu h - \frac{1}{2}m_h^2 h^2 - \frac{1}{4}(F_B)_{\mu\nu}(F_B)^{\mu\nu} + \frac{1}{2}m_A^2 B_\mu B^\mu, \\ m_h = \sqrt{\lambda}\,v, \qquad m_A = qv. \end{gathered} \tag{21.7}$$

The radial field obeys $(\Box + m_h^2)h = 0$. The vector is a Proca field*: it obeys*

$$\partial_\mu (F_B)^{\mu\nu} + m_A^2 B^\nu = 0. \tag{21.8}$$

It satisfies $\partial_\mu B^\mu = 0$ and has three independent polarisations.

Proof. Replace h and B by εh and εB in (21.6) and retain the coefficient of ε^2. The scalar potential contributes $\frac{1}{2}\lambda v^2 h^2$. The vector mass term contributes $\frac{1}{2}q^2v^2 B_\mu B^\mu$. The terms involving hB^2 and h^3 have degree three, so they do not enter the quadratic action. This proves the split and both masses.

Varying the quadratic density with compactly supported variations gives the stated equations by coordinate integration by parts, as in Lecture 9. For the vector, the kinetic variation gives $\partial_\mu(F_B)^{\mu\nu}$ and the mass variation gives $m_A^2 B^\nu$. Taking the divergence, symmetry of mixed derivatives and antisymmetry of F_B give $m_A^2\partial_\nu B^\nu = 0$. Since $m_A > 0$, the divergence vanishes. Equation (21.8) then reduces to $(\Box + m_A^2)B^\nu = 0$.

For a plane wave $B^\mu = \epsilon^\mu e^{-\mathrm{i}\nu t + \mathrm{i}k\cdot x}$, let $k^\mu = (\nu, k)$, where $\nu = \sqrt{|k|^2 + m_A^2} > 0$. The divergence constraint is $\nu\epsilon^0 - k\cdot\epsilon = 0$. It determines ϵ^0 from three arbitrary spatial components. In the rest frame $k = 0$, $\nu = m_A$ and $\epsilon^0 = 0$. The three spatial axes give three independent polarisations. Real solutions are the real parts of these modes. ∎

For $k = (0, 0, \kappa)$, two unit polarisation vectors are the spatial x and y directions. A third, longitudinal one is

$$\epsilon_L^\mu = \frac{1}{m_A}(\kappa, 0, 0, \nu), \qquad k_\mu \epsilon_L^\mu = 0, \qquad \epsilon_{L,\mu}\epsilon_L^\mu = -1.$$

At rest it becomes the third spatial unit vector. The divergence constraint is not a freedom to remove this component by gauge: B itself is gauge invariant.

21.3 The canonical count in the quadratic vector sector

We apply the canonical bracket of Lecture 11 directly to the Proca density. Write $B = -\beta\,\mathrm{d}t + \mathbf{b}\cdot\mathrm{d}x$ and $m = m_A$. For this calculation use Schwartz data and smearings, so face terms vanish. The vector density is

$$L_B = \frac{1}{2}|\mathbf{b}_t + \nabla\beta|^2 - \frac{1}{2}|\operatorname{curl}\mathbf{b}|^2 + \frac{1}{2}m^2(\beta^2 - |\mathbf{b}|^2).$$

The momenta are $\pi_\beta = 0$ and $\mathbf{\Pi}_b = \mathbf{b}_t + \nabla\beta$. On a cube Q_R, integration of $-\mathbf{\Pi}_b\cdot\nabla\beta$ leaves $-\int_{\partial Q_R}\beta\mathbf{\Pi}_b\cdot n\,\mathrm{d}S$ as its boundary term. Schwartz decay makes this tend to zero as $R\to\infty$. The Hamiltonian is

$$H_B = \int\left[\frac{1}{2}|\mathbf{\Pi}_b|^2 + \frac{1}{2}|\operatorname{curl}\mathbf{b}|^2 + \frac{1}{2}m^2|\mathbf{b}|^2 + \beta\operatorname{div}\mathbf{\Pi}_b - \frac{1}{2}m^2\beta^2\right]\mathrm{d}^3x.$$

Preserving $\pi_\beta = 0$ gives $\Gamma_m = \operatorname{div}\mathbf{\Pi}_b - m^2\beta = 0$. For real scalar test functions f and g, set $P[f] = \int f\pi_\beta$ and $C_m[g] = \int g\Gamma_m$. The bracket is

$$\{P[f], C_m[g]\} = m^2\int fg\,\mathrm{d}^3x.$$

The primary–secondary bracket is the invertible multiplication by m^2. This pair is second class, unlike the commuting Maxwell constraints of Lecture 11. Its equations solve for $\pi_\beta = 0$ and $\beta = m^{-2}\operatorname{div}\mathbf{\Pi}_b$. As in Lecture 11, adding $\int\zeta\pi_\beta$ to the Hamiltonian initially leaves $\beta_t = \zeta$ arbitrary. The spatial momentum equation is $\partial_t\mathbf{\Pi}_b = -\operatorname{curl}\operatorname{curl}\mathbf{b} - m^2\mathbf{b}$. Preserving Γ_m therefore fixes $\beta_t = -\operatorname{div}\mathbf{b}$, rather than imposing another spatial constraint.

Pulling the canonical form to these constraint values leaves $\int\delta\mathbf{b}\wedge\delta\mathbf{\Pi}_b$: the scalar term contains $\delta\pi_\beta = 0$. Thus all three spatial canonical pairs remain. This is the constraint count of the quadratic Proca sector, computed here. Together with the radial scalar it gives $3 + 1 = 4$ linearised modes. The uncoupled description has the same count: two Maxwell modes from Lecture 12 and two real scalar modes from Lecture 17.

21.4 The zero-coupling limit in the original variables

Example 21.4 (Returning to the global scalar model)**.** Keep $v > 0$ and $\lambda > 0$ fixed and let $q \downarrow 0$. Before introducing B, the quadratic density is

$$L^{(2)} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\partial_\mu h\partial^\mu h + \frac{1}{2}v^2(\partial\theta - qA_{\mathrm{em}})^2 - \frac{1}{2}\lambda v^2h^2.$$

The squared one-form denotes contraction with $\eta^{\mu\nu}$. Setting $q = 0$ gives a free Maxwell field, the radial scalar of mass $\sqrt{\lambda}\,v$, and the massless scalar $\xi = v\theta$. This is the canonically normalised phase of Lecture 17. The count is now $2 + 1 + 1 = 4$.

The expression in $(A_{\mathrm{em}}, \theta)$ is regular at $q = 0$. The replacement $B = A_{\mathrm{em}} - q^{-1}\mathrm{d}\theta$ is singular there, so it cannot be used to take that limit with θ fixed. For $q > 0$, the phase contributes to the massive vector's longitudinal mode. At $q = 0$, it is an independent massless scalar again.

21.5 A vortex zero obstructs a smooth unitary gauge

A nonzero complex field has a real phase only locally in general. Around a zero, phase winding can prevent a global branch. Consider the local planar model $\Phi(x, y) = x + \mathrm{i}y$. On a circle about the origin its phase winds once, as shown in Figure 21.1. This is a model of the zero, not a claim of finite energy for that profile on the whole plane.

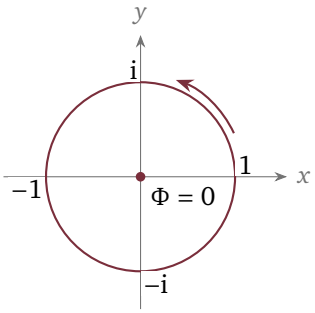


Figure 21.1: For $\Phi = x + \mathrm{i}y$, the field phase winds once around the zero. The four labels show its values on the unit circle.

On the punctured plane the circle-valued gauge function

$$g(x, y) = \frac{x - \mathrm{i}y}{\sqrt{x^2 + y^2}}$$

is smooth and makes $g\Phi = \sqrt{x^2 + y^2}$ positive. On each phase patch it is $e^{\mathrm{i}q\chi}$ with $\chi = -\arg(x + \mathrm{i}y)/q$. It has no single-valued real logarithm on the punctured plane, since its phase winds once around the origin. Any unit-modulus function u with $u\Phi \in \mathbb{R}_{>0}$ must equal $\overline{\Phi}/|\Phi| = g$, so no alternative unitary gauge avoids this obstruction. The function g cannot extend continuously through the zero: along the positive and negative x axes its limits are 1 and -1. The associated potential transformation contains $-q^{-1}\mathrm{d}\arg(x + \mathrm{i}y)$, which is singular at the origin. Consequently a circle-valued unitary gauge on the punctured plane does not supply a smooth unitary gauge through a vortex zero. More generally, a smooth circle-valued extension through a zero is contained in a small phase arc on a sufficiently small disk, by continuity. It therefore has a real phase there and zero winding on small circles. At a vortex zero with nonzero phase winding, the forced unitary gauge has the opposite nonzero winding, a contradiction. Lecture 22 keeps the phase winding and the core rather than applying (21.3) globally.

Summary. With kinetic coefficient $1/2$ and potential coefficient $\lambda/8$, the local Higgs expansion gives radial mass $\sqrt{\lambda}\,v$ and vector mass qv. The gauge-invariant vector obeys Proca's divergence constraint and has three polarisations. Its quadratic constraint pair is second class, leaving three spatial canonical pairs and one radial scalar mode. The original phase variables have a regular $q \to 0$ limit with a massless phase. A vortex zero prevents extending a local unitary gauge smoothly through its core.

Self-check. For $q = 2$, $v = 3$ and $\lambda = 9$, compute the two masses. Which coefficient in the scalar potential fixes the radial value?

Answer. The vector mass is $qv = 6$, and the radial mass is $\sqrt{\lambda}\, v = 9$. The expansion of $(\lambda/8)(2vh + h^2)^2$ has quadratic term $\frac{1}{2}\lambda v^2 h^2$.

Self-check. Under $\theta \mapsto \theta + q\chi$ and $A_{\mathrm{em}} \mapsto A_{\mathrm{em}} + \mathrm{d}\chi$, compute the changes in $\mathrm{d}\theta - qA_{\mathrm{em}}$ and in B.

Answer. Both changes vanish. The two added terms in $\mathrm{d}\theta - qA_{\mathrm{em}}$ are $q\mathrm{d}\chi$ and $-q\mathrm{d}\chi$. The cancellation in B is $\mathrm{d}\chi - q^{-1}q\mathrm{d}\chi = 0$.

Self-check. Which variables exhibit the regular $q = 0$ limit, and what replaces the third massive-vector polarisation in that description?

Answer. The variables $(A_{\mathrm{em}}, \theta)$ are regular. The massless Maxwell field has two polarisations, and $\xi = v\theta$ supplies the independent massless scalar. The radial mode remains massive.

Self-check. Does $g = (x - \mathrm{i}y)/\sqrt{x^2 + y^2}$ provide a smooth unitary gauge on the whole plane for $\Phi = x + \mathrm{i}y$?

Answer. It works only on the punctured plane. Its limits from the two x directions differ, so it has no continuous extension to the zero. The local polar decomposition cannot be used there.